Systematic ε-Greedy: A Deterministic Periodic Exploration Schedule for Multi-Armed Bandits

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***Abstract*—The standard way to implement ε-greedy exploration is to explore whenever a uniform random draw falls below ε, which makes both the number and the timing of exploratory pulls random. We propose Systematic ε-greedy, a deterministic alternative that explores on a fixed clock, once every $S = \lfloor 1/\varepsilon \rfloor$ steps, and a phase-randomized extension in which each run starts its clock at a random offset. We evaluate both against standard stochastic ε-greedy on a stationary 10-armed Gaussian testbed ($\sigma \in \{0.1, 1, 2\}$, $\varepsilon \in \{0.1, 0.01\}$, $T = 2{,}000$) using 4,000 paired environments per configuration. The basic fixed-phase method matches the standard method on final regret (0.6–3.1% lower, none beyond 2 standard errors). The phase-randomized version lowers final regret by 6–10% at $\varepsilon = 0.01$ and by 1–2% at $\varepsilon = 0.1$, and the benefit does not grow with noise. About half of the gain at low ε comes from removing variance in the number of explorations, according to a control experiment. We also show that the apparent superiority of the basic method in raw per-step curves is a measurement artifact of synchronized exploration, and we give an evaluation protocol that avoids it. The method was developed independently; a limited literature search found related derandomized and forced-exploration schemes but no direct evaluation of this exact schedule.**

I. Introduction

An agent facing the exploration–exploitation trade-off must keep choosing between its best current estimate and alternatives it knows less about. The usual ε-greedy implementation draws a random number at every step and explores when it falls below ε. Because each step is an independent coin flip, the agent can explore several times in a row, go a long time without exploring, and end up with a different total number of explorations in every run.

We propose replacing the coin flip with a clock. If $\varepsilon = 1/S$ for an integer S, explore on every S-th step and exploit otherwise. This Systematic ε-greedy schedule has the same exploration rate as the standard method, uses an exact exploration budget (T/S pulls), needs no random draw to decide when to explore, and makes the exploration timing reproducible and easy to audit.

Whether this is actually better is an empirical question, and the answer turns out to be subtler than we first expected. Our contributions are:

- **A method.** Systematic ε-greedy (fixed-phase) and its phase-randomized extension.
- **An honest evaluation.** A paired comparison with standard ε-greedy over six configurations: the basic method ties, and the phase-randomized version gives a modest, statistically clear gain at low ε.
- **An evaluation pitfall.** With synchronized exploration, per-step curves alternate between exploit and explore steps and can look better than they are.
- **A mechanism study.** A control experiment separating the effect of the number of explorations from their spacing.

II. Related work and positioning

ε-greedy and the 10-armed testbed follow Sutton & Barto (2018, Chapter 2). Common alternatives are ε-first / explore-first, in which a block of rounds at the start is dedicated to exploration (Slivkins, 2019), decaying ε, UCB (Auer, Cesa-Bianchi & Fischer, 2002), and Thompson sampling (Thompson, 1933).

Two lines of work are close to ours. First, Even-Dar & Mansour (2001) introduce incremental Q-learning, which gradually promotes the values of actions that are not taken and is described by its authors as a derandomization of ε-greedy Q-learning. That scheme is value-driven and set in MDPs, whereas ours is a fixed, history-independent clock in the bandit setting. Second, forced-exploration methods for bandits (see "Forced Exploration in Bandit Problems", arXiv:2312.07285) interleave greedy pulls with forced exploration steps, but their exploration schedule adapts to the history of pulls, whereas ours does not.

Scope of the novelty claim. The idea was developed independently. In a limited, non-systematic search we did not find a direct evaluation of a fixed-period ε-greedy schedule ($t \bmod \lfloor 1/\varepsilon \rfloor = 0$, with uniformly random arm choice) against Bernoulli ε-greedy in stationary bandits, but this does not rule out prior work. We therefore claim the method, the evaluation and the analysis here, not priority.

III. Proposed method

Standard baseline (Stochastic ε-greedy). At each step, with probability ε pick a uniformly random arm; otherwise pick $\arg\max_a Q(a)$.

Systematic ε-greedy (fixed-phase). Let $S = \lfloor 1/\varepsilon \rfloor$.

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for t = 0, 1, ..., T-1:
   if t mod S == 0: a = uniformly random arm #
       explore
   else: a = argmax_a Q(a) # exploit
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Phase-randomized Systematic ε-greedy. Draw an offset $\varphi \sim \mathrm{Uniform}\{0, \ldots, S-1\}$ once per run and explore when $(t + \varphi) \bmod S = 0$. The number of explorations is the same, but different runs no longer explore on the same steps.

Properties and caveats.

- The exploration rate is exactly $1/S$, so ε values for which $1/\varepsilon$ is an integer are used (0.1 and 0.01 here).
- The exploration count is deterministic (T/S), while for the standard method it is $\mathrm{Binomial}(T, \varepsilon)$.
- Fixed-phase has two quirks. All runs explore on the same steps, which creates a synchronization artifact in averaged curves (Section 5.3). And at $t = 0$ all estimates are equal, so the exploration at $t = 0$ is redundant, which wastes one exploration (one of 20 at $\varepsilon = 0.01$). Phase randomization addresses both.

IV. Experimental setup

A. Environment

A stationary bandit with $k = 10$ arms. For each run, true values are drawn as $q(a) \sim N(5, 1)$; pulling arm a returns a reward from $N(q(a), \sigma^2)$, with $\sigma \in \{0.1, 1.0, 2.0\}$.

B. Agent

Estimates use the incremental sample average, $Q_{n+1} = Q_n + (1/n)(R_n - Q_n)$, with $Q_0 = 0$. Because rewards are around 5, this is pessimistic initialization: after an arm has been pulled, the greedy rule never tries an untried arm, so all discovery of new arms comes from exploration. This makes exploration schedules matter more than they would with optimistic initialization. Ties in argmax go to the lowest index.

C. Evaluation protocol

All strategies run on the same environments (same seeds), so comparisons are paired. The figures use 500 environments (seed 0) with ± 1 standard-error ribbons (SD / $\sqrt{500}$). The headline table uses 8 seeds $\times$ 500 = 4,000 environments per configuration, because a single 500-environment run is noisy (Section 5.4). Metrics are final cumulative pseudo-regret at $T = 2{,}000$ (computed from true arm values), the paired difference from Stochastic $\pm$ standard error, the mean % optimal action over the last 500 steps, and the mean reward over the last 500 steps.

V. Results

A. Final cumulative regret (4,000 paired environments)

Δ is the paired difference from Stochastic (negative = lower regret) $\pm$ standard error. Bold marks Δ beyond 3 standard errors. All six configurations use the same environments, so rows are not independent replications.

The basic fixed-phase method is not better or worse in any practical sense. Fixed-phase differences are all slightly negative (0.6–3.1% lower regret) but none exceeds 2 standard errors.

Random-phase gives a modest, clear gain at low ε: 9.5% ($\sigma = 0.1$), 7.5% ($\sigma = 1$) and 5.8% ($\sigma = 2$) lower regret at $\varepsilon = 0.01$, and about 2.3%, 2.3% and 1.1% at $\varepsilon = 0.1$ (significant for $\sigma = 0.1$ and 1 only).

Noise does not help. The relative benefit shrinks as σ grows, so our initial hypothesis that deterministic spacing helps most in noisy environments is not supported.

ε matters far more than the schedule. $\varepsilon = 0.1$ gives roughly half the regret of $\varepsilon = 0.01$ at $T = 2{,}000$, because under pessimistic initialization 20 explorations across 10 arms are too few to reliably find the best arm.

B. Late-stage performance

Mean % optimal action and mean reward over the last 500 steps are essentially identical across strategies at $\varepsilon = 0.1$ (for example at $\sigma = 0.1$: 90.0%, 90.3%, 90.1%; at $\sigma = 2$: 75.6%, 75.5%, 75.5%). At $\varepsilon = 0.01$ both periodic schedules are about 1.3–2.3 points higher in % optimal and about 0.02 higher in average reward (borderline to moderately significant, depending on σ). The effect on reward is tiny in absolute terms (about 0.3%).

C. Figures

Each figure shows average reward, % optimal action and cumulative regret for 500 environments (seed 0). Red is Stochastic, blue is fixed-phase Systematic (the basic method), green is random-phase Systematic.

Reading the $\varepsilon = 0.1$ figures (1, 3, 5). The blue block is not a thick line. Because all runs explore on the same steps, the averaged fixed-phase curve alternates every 10 steps between the exploit-step value (high) and about $1/k = 10\%$ on exploration steps, which fills the block and hides the other curves. Its upper edge sits above Stochastic's curve, but Stochastic's curve is already a mixture of explore and exploit steps. If exploit-step accuracy is 95%, the fixed-phase period average is $0.9 \times 95\% + 0.1 \times 10\% \approx 86.5\%$. Compared fairly, as in the regret panel and Section 5.2, the strategies are close. A naive reading of the raw curves would suggest that Systematic identifies the optimal arm more often at moderate to high noise; the aggregate metrics do not support that.

Reading the $\varepsilon = 0.01$ figures (2, 4, 6). The fixed-phase curve is a staircase with downward spikes every 100 steps: it rises only at exploration steps, when a better arm can be found, and drops to about 10% on those steps. Between spikes it is not near 100% (the last-500-step mean is roughly 83%, 67% and 55% for $\sigma = 0.1$, 1 and 2). The random-phase curve is smooth and tends to lead in the first half of the run. Regret is still concave at $T = 2{,}000$ for $\varepsilon = 0.01$, so "regret grows linearly" holds only for $\varepsilon = 0.1$.

D. Single-seed figures overstate the effect

In the 500-environment, seed-0 figures, random-phase at $\varepsilon = 0.01$ shows regret reductions of about 116, 115 and 123 ($\sigma = 0.1$, 1, 2). With 4,000 environments these shrink to about 74, 69 and 66. The seed-0 numbers are noisy estimates (standard errors around 40–50), and reporting them alone would overstate the gain by roughly 60–85%. We therefore report the 4,000-environment table as the main result and the figures as illustrations.

VI. Why does random phase help?

Counts, not order. With sample-average estimates in a stationary environment, an arm's estimate depends on how

σ	ε	Stochastic	Fixed-phase Δ	Random-phase Δ
0.1	0.1	385.3 ± 2.5	-2.3 ± 1.9	-8.8 ± 1.8
0.1	0.01	773.6 ± 12.7	-23.9 ± 15.1	-73.5 ± 13.7
1.0	0.1	438.3 ± 2.9	-6.0 ± 3.1	-10.2 ± 3.0
1.0	0.01	922.5 ± 13.7	-26.1 ± 17.0	-69.0 ± 15.3
2.0	0.1	538.4 ± 4.5	-6.3 ± 5.7	-6.1 ± 5.8
2.0	0.01	1129.0 ± 15.7	-23.4 ± 19.8	-65.8 ± 17.7

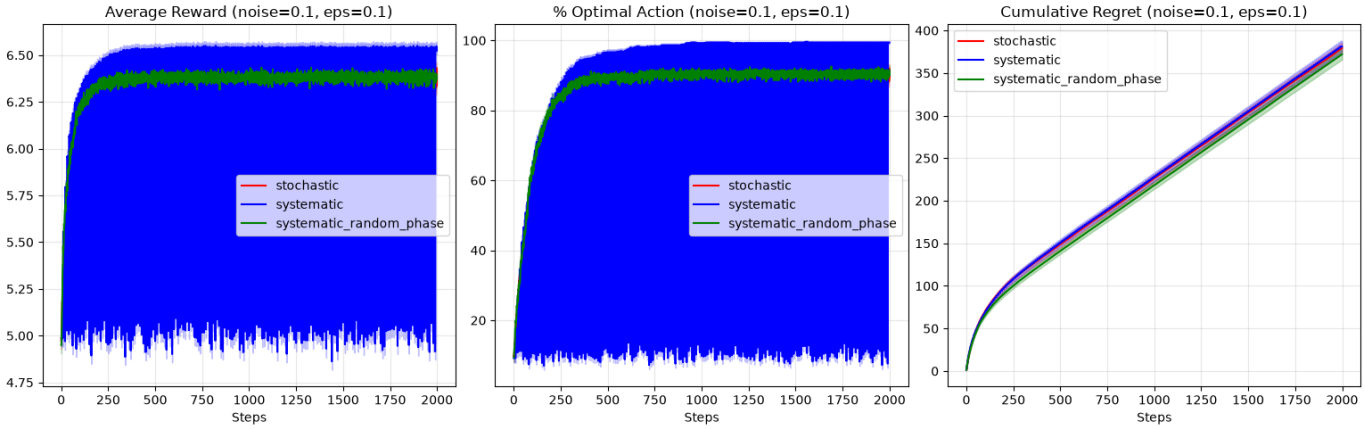


Fig. 1. $\sigma = 0.1$, $\varepsilon = 0.1$

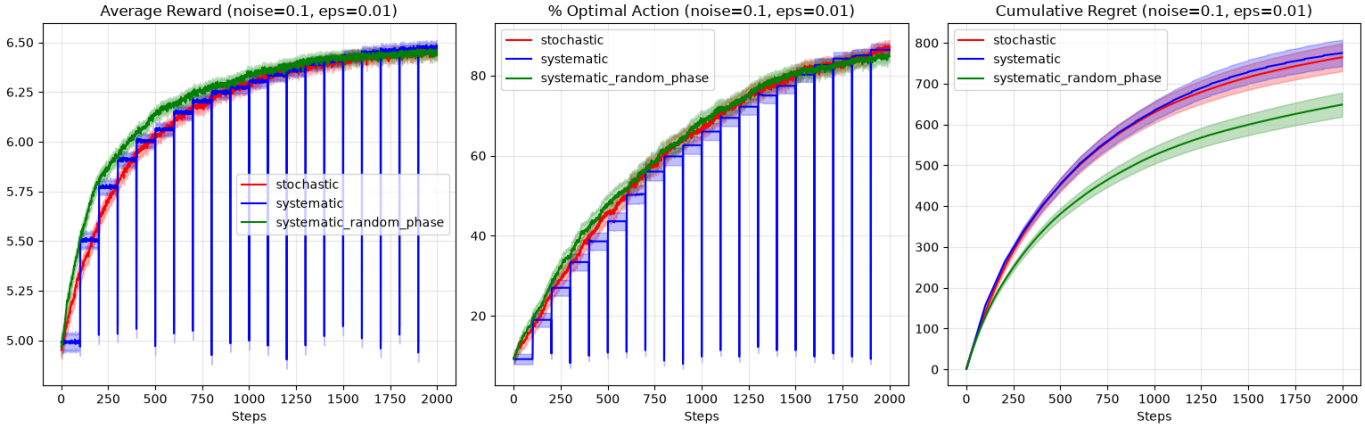


Fig. 2. $\sigma = 0.1$, $\varepsilon = 0.01$

many times it was pulled and the rewards it returned, not on how pulls were spaced.

Same mean, less variance. Random-phase and Stochastic have the same expected number of exploratory pulls at every step, but Stochastic's count is binomial: at $T = 2{,}000$, SD ≈ 4.5 around 20 at $\varepsilon = 0.01$ (22% relative) and ≈ 13.4 around 200 at $\varepsilon = 0.1$ (7%). If regret is convex and decreasing in the number of explorations, more variance means higher expected regret, with the largest effect when explorations are few. This matches the larger gain at $\varepsilon = 0.01$.

Control experiment. We added a control with exactly εT explorations per run placed at uniformly random steps (fixed count, random timing). Its regret difference from Stochastic is:

At $\varepsilon = 0.01$, fixing the count alone recovers roughly 50–60% of the random-phase gain. The remainder (about 25–35 regret units) plausibly comes from even spacing, though we did not test that gap separately. At $\varepsilon = 0.1$ the effects are small throughout.

Why fixed-phase gains less. It loses one exploration to the $t = 0$ redundancy and has its first real exploration at step S rather than at a uniformly random time in $[0, S)$, so on average it is half an exploration behind at any step. At $\varepsilon = 0.01$ this offsets much of the variance benefit.

VII. Limitations

1) Stationary, pessimistically initialized bandit only. Rerun with $Q_0 =$ optimistic or neutral; effect sizes may change

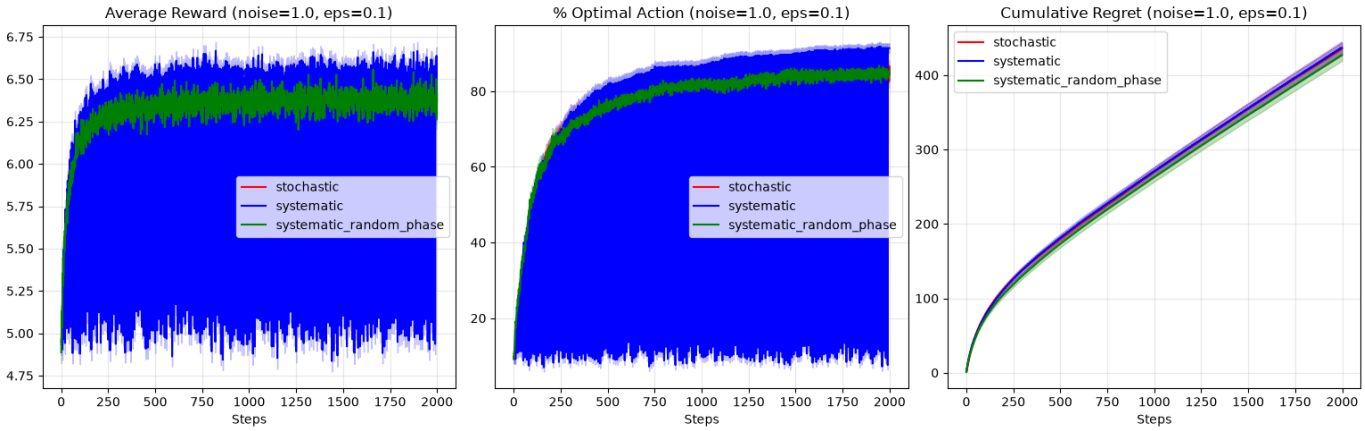


Fig. 3. $\sigma = 1$, $\varepsilon = 0.1$

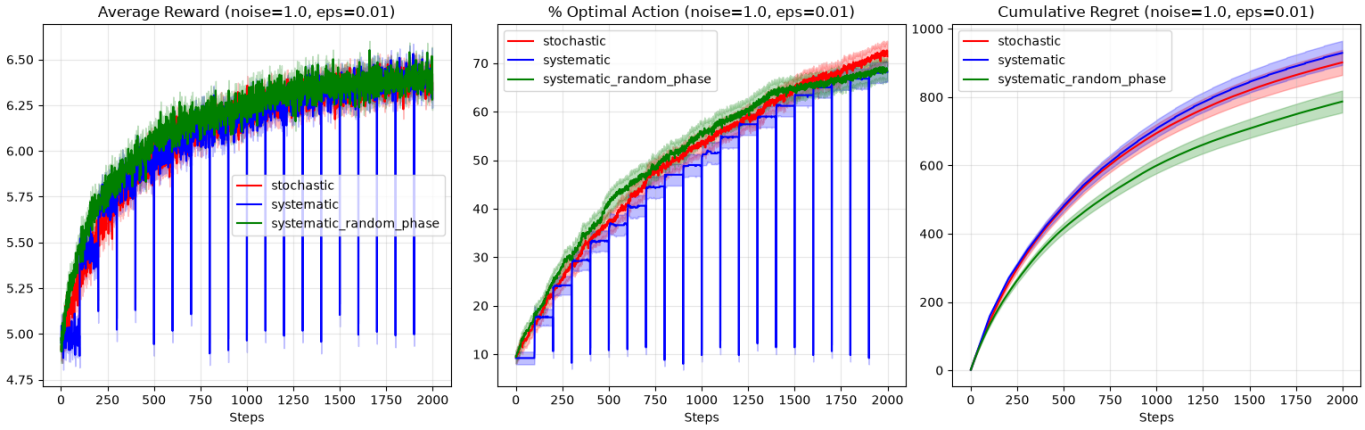


Fig. 4. $\sigma = 1$, $\varepsilon = 0.01$

σ	ε	Fixed count, random timing Δ	Random-phase Δ
0.1	0.01	-38.3 ± 13.9	-73.5 ± 13.7
1.0	0.01	-39.7 ± 15.5	-69.0 ± 15.3
2.0	0.01	-40.6 ± 18.0	-65.8 ± 17.7
0.1 / 1.0 / 2.0	0.1	-1.1 ± 1.8 / -3.5 ± 3.0 / -10.4 ± 5.6	-8.8 / -10.2 / -6.1

substantially.

2) One horizon and two ε values. Try $T \geq 10{,}000$, $k \in \{50, 100\}$, and more ε values.
3) Not independent across rows. The six configurations share environments; replicate with different seeds per configuration.
4) Mechanism is partly tested. The count control supports about half the effect; the spacing contribution is an inference.
5) Non-stationary problems. With a constant step size the spacing of samples matters differently; this is the most promising place for a periodic schedule to behave differently.
6) Baselines. Compare against decaying ε, optimistic initialization, UCB and Thompson sampling.

VIII. Conclusion

We proposed Systematic ε-greedy, a clock-driven replacement for the coin flip in ε-greedy. In a stationary Gaussian 10-armed bandit, the basic fixed-phase method is statistically indistinguishable from the standard method on final regret, and its apparent superiority in raw per-step curves is an artifact of synchronized exploration. Giving each run a random phase offset, a one-line change, yields a modest and statistically clear improvement (6–10% lower regret at $\varepsilon = 0.01$, 1–2% at $\varepsilon = 0.1$) that does not grow with noise. Roughly half of the low-ε gain comes from removing variance in the number of exploratory pulls; the rest plausibly comes from even spacing. The practical recommendation is: if you want deterministic exploration, randomize the phase, and evaluate with aggregate metrics, paired environments and standard-

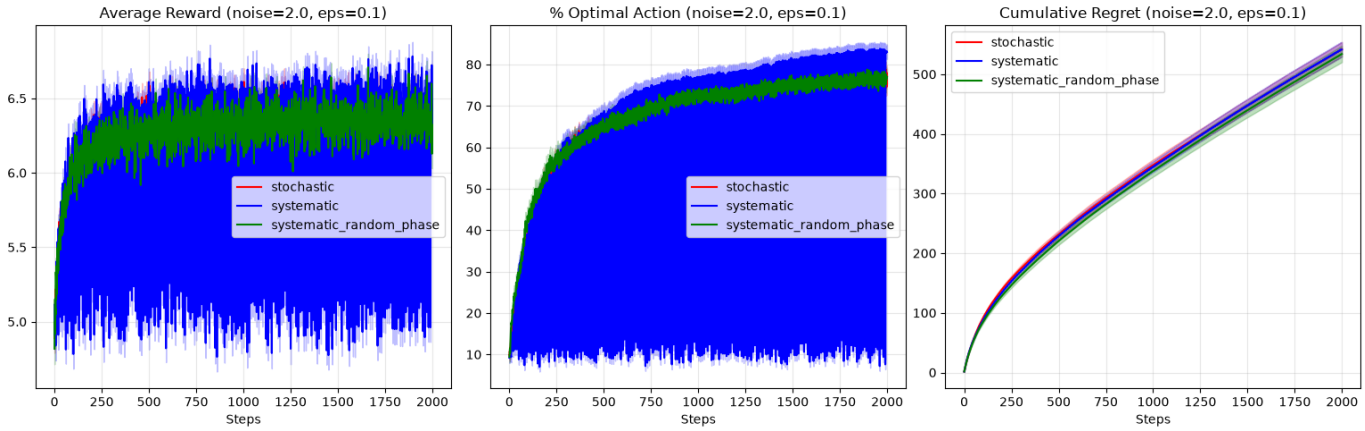


Fig. 5. $\sigma = 2$, $\varepsilon = 0.1$

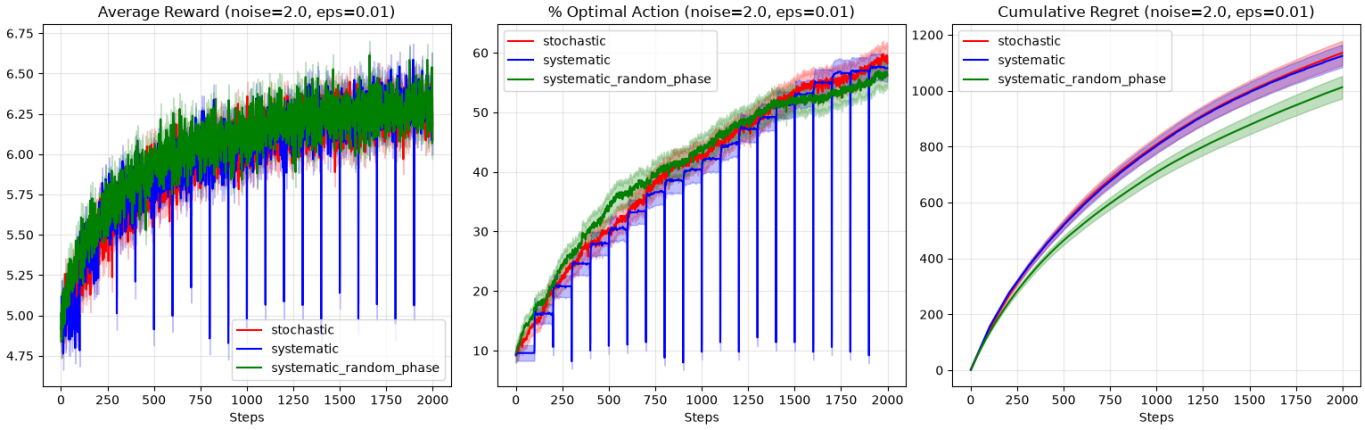


Fig. 6. $\sigma = 2$, $\varepsilon = 0.01$

error intervals. Open questions are non-stationary problems, other initializations, and longer horizons.

Reproducibility

Code: bandit_final.py (500 environments, seed 0, produces the six figures). The 4,000-environment table calls the same simulate function over eight seeds (2000, 2010, …, 2070) and takes paired differences of final regret across environments. The fixed-count control draws exactly εT exploration steps per run uniformly at random.

References

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